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Q5 DECLARE VExpected, i, Cap-A, Cap-B, Cap-C : INTEGER
DECLARE OccA, OccB, OccC, Processed, Allotted : INTEGER
DECLARE Rejected, CarsP, BikesP, VansP : INTEGER.
DECLARE Spaces : INTEGER.
DECLARE Zone : CHAR
DECLARE VType, Ucat, Permit, Emergency : CHAR.
DECLARE Provided : BOOLEAN
DECLARE Reason, Highest : STRING.
VType ← ' ', Ucat ← ' ', Permit ← ' ', Emergency ← ' '
Cap-A ← 20, Cap-B ← 40, Cap-C ← 15
OccA ← 0, OccB ← 0, OccC ← 0.
Processed ← 0, Allotted ← 0, Rejected ← 0
CarsP ← 0, BikesP ← 0, VansP ← 0
INPUT VExpected
FOR i ← 1 to VExpected.
    WHILE VType <> 'C' AND VType <> 'B' AND VType <> 'V'
        OUTPUT "Enter Vehicle Type"
        INPUT VType.
    END WHILE
    WHILE Ucat <> 'P' AND Ucat <> 'S' AND Ucat <> 'G'
        OUTPUT "Enter User Category"
        INPUT Ucat
    END WHILE

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WHILE Permit <> 'Y' AND Permit <> 'N'  
    OUTPUT "Enter Permit Valid or not"  
    INPUT Permit  
ENDWHILE
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IF Permit = 'N' THEN  
    WHILE Emergency <> 'Y' AND Emergency <> 'N'  
        OUTPUT "Enter Emergency or not"  
        INPUT Emergency  
    ENDWHILE  
ENDIF
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Processed ← Processed + 1  
Provided ← FALSE  
Spaces ← 1
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IF Permit = 'N' AND Emergency = 'N' THEN  
    OUTPUT "Rejected as has No Valid Permit  
    Reason ← and is not an Emergency Vehicle"  
    PRINT
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ELSE
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    IF VCal = 'F' THEN
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        Zone ← 'A'
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        Spaces ← 1
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    ELSE IF VCal = 'S' THEN
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```
        IF VType = 'V' THEN
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```
            Zone ←
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        Spaces  $\leftarrow$  2.
    ELSE
        Zone  $\leftarrow$  'B'
        Spaces  $\leftarrow$  1
    ENDIF
ELSE
    Zone  $\leftarrow$  'C'
    IF VType = 'V' THEN
        Spaces  $\leftarrow$  2
    ELSE
        Spaces  $\leftarrow$  1
    ENDIF
ENDIF

IF Zone = 'A' THEN
    IF (CapA - OccA)  $\geq$  Spaces THEN
        OccA  $\leftarrow$  OccA + Spaces
        Provided  $\leftarrow$  TRUE
        OUTPUT "Assigned to Zone A and
            Remaining Spaces are" (CapA - OccA)
    ELSE
        Reason  $\leftarrow$  "No space in Zone A"
    ENDIF
ELSE IF Zone = 'B' THEN
    IF (CapB - OccB)  $\geq$  Spaces THEN
        OccB  $\leftarrow$  OccB + Spaces
        Provided  $\leftarrow$  TRUE
        OUTPUT
    ENDIF

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OUTPUT "Assigned to Zone A and
Remaining spaces are", ~~CapB - OccB~~

ELSE

~~Reason~~ Reason $\leftarrow$ "No Space in Zone B"

ENDIF

ELSE

IF $(CapC - OccC) \geq \text{Spaces}$ THEN

Provided $\leftarrow$ TRUE

$OccC \leftarrow OccC + \text{Spaces}$

OUTPUT "Assigned to Zone C and
Remaining Spaces are",
 $(CapB - OccB)$

ELSE

Reason $\leftarrow$ "No Space in Zone C"

ENDIF

ENDIF

ENDIF

IF Provided = TRUE THEN

Allotted $\leftarrow$ Allotted + 1

IF VType = 'C' THEN

CarsP $\leftarrow$ CarsP + 1

ELSEIF VType = 'B' THEN

BikesP $\leftarrow$ BikesP + 1

ELSE

VansP $\leftarrow$ VansP + 1

ENDIF

ELSE

Rejected $\leftarrow$ Rejected + 1

OUTPUT "The Vehicle ~~is~~ cannot be.
Allotted a parking space
Because", Reason.

ENDIF

NEXT

OUTPUT "Parking Summary"

OUTPUT "Total Processed Vehicles:", Processed

OUTPUT "Total Accepted Vehicles:", Provided

OUTPUT "Total ~~Acc~~ Rejected Vehicles:", Rejected.

OUTPUT "Cars Parked:", CarsP.

OUTPUT "Bikes Parked:", BikesP

OUTPUT "Vans Parked:", VansP

OUTPUT "ZoneA Occupancy:", OccA

OUTPUT "Zone A Remaining Capacity:", (CapA - OccA)

OUTPUT "Zone B ~~Acc~~ Occupancy:", OccB

OUTPUT "Zone B Remaining Capacity:", (CapB - OccB)

OUTPUT "Zone C Occupancy:", OccC

OUTPUT "Zone C Remaining Capacity:", (CapC - OccC)

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IF OccA > OccB AND OccA > OccC THEN
    OUTPUT "Zone A has highest Occupancy"
ELSE IF OccB > OccA AND OccB > OccC THEN
    OUTPUT "Zone B has highest Occupancy"
ELSE IF OccC > OccA AND OccC > OccB THEN
    OUTPUT "Zone C has highest Occupancy"
ELSE IF OccA = OccB AND OccA > OccC THEN
    OUTPUT "Zone A & B have Highest Occupancy"
ELSE IF OccA = OccC AND OccB < OccA THEN
    OUTPUT "Zone A & C have Highest Occupancy"
ELSE IF OccB = OccC AND OccB > OccA THEN
    OUTPUT "Zone B & C have Highest Occupancy"
ELSE {
    OUTPUT "All have equal Occupancy"
ENDIF

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IF (OccA + OccB + OccC) = (CapA + CapB + CapC) THEN
    OUTPUT "The Parking at Campus is full"
ELSE
    OUTPUT "The Parking at Campus is not full"
ENDIF

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